

1995 HKCEE MATHS Paper II

- 1 Round off the number 0.044449 to 3 significant figures.

A. 0.04
 B. 0.044
 C. 0.045
 D. 0.0444
 E. 0.0445

- 2 If $\frac{x+y}{xy} = 1$, then $y =$

A. $\frac{1-x}{x}$
 B. $\frac{x-1}{x}$
 C. $\frac{x}{1-x}$
 D. $\frac{x}{x-1}$
 E. $\frac{1-x}{1+x}$

- 3 If $f(x) = x^{99} + 99x + k$ is divisible by $x + 1$, then $k =$

A. -100
 B. -98
 C. 98
 D. 100
 E. 198

- 4 Simplify $\left(\frac{a^6}{b^{12}}\right)^{-\frac{2}{3}}$

A. $\frac{b^8}{a^4}$
 B. $\frac{b^{18}}{a^9}$
 C. $\frac{a^4}{b^8}$
 D. $\frac{a^9}{b^{18}}$
 E. $\frac{1}{a^4 b^{12}}$

- 5 $\frac{1}{2+\sqrt{6}} - \frac{1}{2-\sqrt{6}} =$

A. $-\sqrt{6}$
 B. $-\frac{\sqrt{6}}{2}$
 C. 0
 D. $\frac{\sqrt{6}}{2}$
 E. $\sqrt{6}$

- 6 The L.C.M. of $x^3 - x$ and $x^4 - 1$ is

A. $x-1$
 B. $(x-1)(x+1)$
 C. $x(x-1)(x+1)(x^2+1)$
 D. $(x-1)(x+1)(x^2+1)(x^2+x+1)$
 E. $x(x-1)^2(x+1)^2(x^2+1)$

7

Solve the simultaneous equations:
$$\begin{cases} 4x - \frac{y}{3} = 6 \\ 2x + \frac{y}{6} = -1 \end{cases}$$

A. $x = -\frac{1}{2}, y = -12$
 B. $x = -\frac{1}{2}, y = 12$
 C. $x = \frac{1}{2}, y = -12$
 D. $x = \frac{1}{2}, y = 12$
 E. $x = \frac{5}{24}, y = -\frac{7}{2}$

- 8 Which of the following shaded regions

represents the solution of
$$\begin{cases} y \geq 0 \\ x - y \geq -3 \\ x + 2y \leq 0 \end{cases}$$

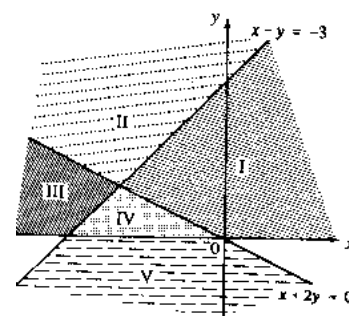
A. I

B. II

C. III

D. IV

E. V



- 9 Find the values of x which satisfy both $-x < 4$

and $\frac{2x-16}{3} > -2$

- A. $-4 < x < 5$
 B. $x < -4$ D. $x < 5$
 C. $x > -4$ E. $x > 5$
- 10 If $3x^2 + 6x + 1 \equiv 3(x+b)^2 + c$, then $c =$

- A. -8
 B. -2 D. $\frac{1}{3}$
 C. 0 E. 1

- 11 x and y are two variables. The table below shows some values of x and their corresponding values of y .

x	2	3	6	12
y	36	16	4	1

Which of the following may be a relation between x and y ?

- A. $x \propto \sqrt{y}$
 B. $x \propto y$ D. $x \propto \frac{1}{y}$
 C. $x \propto \frac{1}{\sqrt{y}}$ E. $x \propto \frac{1}{y^2}$
- 12 If $125^x = 25^y$ and x, y are non-zero, find $x : y$.

- A. $1 : 25$
 B. $1 : 5$ D. $3 : 2$
 C. $2 : 3$ E. $5 : 1$
- 13 Find the interest on $\$P$ at $r\%$ p.a. for n years, compounded half-yearly.

A. $\$P(1+2r\%)^n - \P

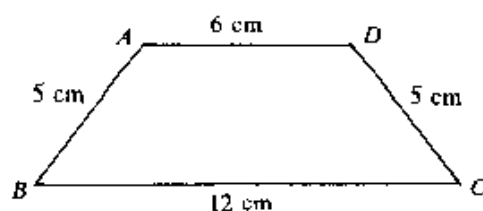
B. $\$P(1+r\%)^n - \P

C. $\$P(1+r\%)^{2n} - \P

D. $\$P\left(1+\frac{r}{2}\%\right)^n - \P

E. $\$P\left(1+\frac{r}{2}\%\right)^{2n} - \P

- 14 In the figure, $ABCD$ is a trapezium. Find its area.



- A. 36cm^2
 B. 45cm^2 D. 72cm^2
 C. 48cm^2 E. 90cm^2
- 15 In the figure, the solid consists of a cylinder and a right circular cone with a common base which is a circle of radius 3cm. The height of the cylinder is 10cm and the slant height of the cone is 5cm. Find the total surface area of the solid.

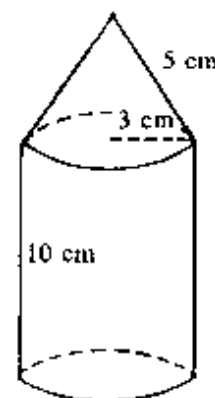
A. $75\pi\text{cm}^2$

B. $84\pi\text{cm}^2$

C. $93\pi\text{cm}^2$

D. $105\pi\text{cm}^2$

E. $114\pi\text{cm}^2$



16 $\frac{\cos^2 \theta}{1 + \sin \theta} - 1 =$

A. $-\sin \theta$

B. $\sin \theta$

D. $-\frac{\sin \theta(1 - \sin \theta)}{1 + \sin \theta}$

C. $\sin \theta - 2$ E. $\frac{\sin \theta(1 - \sin \theta)}{1 + \sin \theta}$

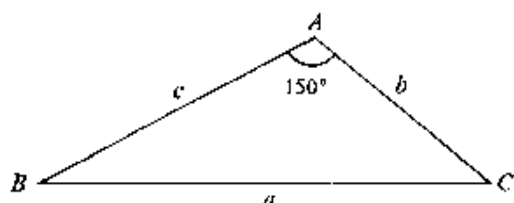
- 17 If $0 < x < 2\pi$, solve $\sin x = \frac{1}{3}$ correct to 3 significant figures.

- A. 0.327 or 2.81
 B. 0.327 or 3.47 D. 0.340 or 3.48
 C. 0.340 or 2.80 E. 0.340 or 5.94

- 18 The greatest value of $\frac{1}{2^{1-\sin x}}$ is

- A. $\frac{1}{2}$
 B. $\frac{1}{4}$ D. 2
 C. 1 E. 4

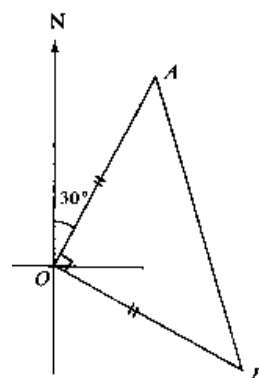
- 19 According to the figure, which of the following must be true?



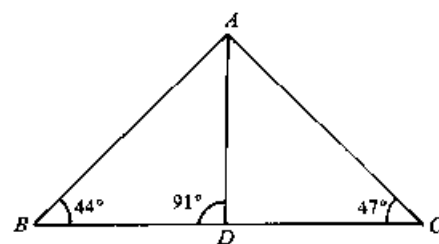
- A. $a^2 = b^2 + c^2 - \sqrt{3}bc$
 B. $a^2 = b^2 + c^2 - bc$
 C. $a^2 = b^2 + c^2 + \frac{\sqrt{3}}{2}bc$
 D. $a^2 = b^2 + c^2 + bc$
 E. $a^2 = b^2 + c^2 + \sqrt{3}bc$

- 20 In the figure, the bearing of B from A is

- A. 015°
 B. 045°
 C. 075°
 D. 165°
 E. 345°



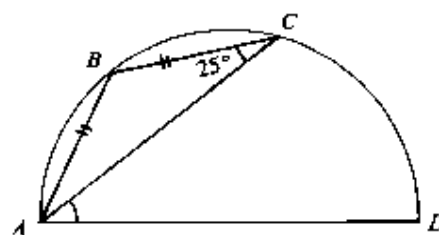
- 21 In the figure, BDC is a straight line. Arrange AD , BD and DC in ascending order of magnitude.



- A. $AD < BD < DC$
 B. $AD < DC < BD$
 C. $DC < AD < BD$
 D. $DC < BD < AD$
 E. $BD < AD < DC$

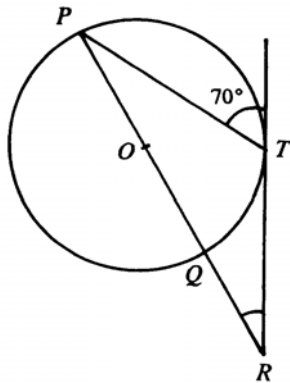
- 22 In the figure, $ABCD$ is a semicircle. $\angle CAD =$

- A. 25°
 B. 40°
 C. 45°
 D. 50°
 E. 65°

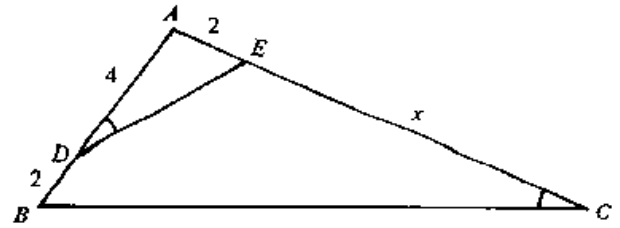


- 23 In the figure, O is the center of the circle, $POQR$ is a straight line. TR is the tangent to the circle at T . $\angle PRT =$

- A. 20°
 B. 35°
 C. 45°
 D. 50°
 E. 70°



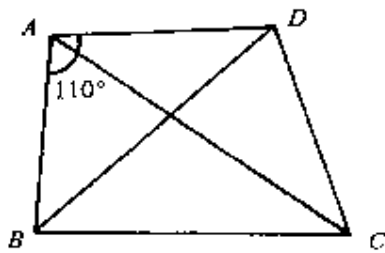
- 26 In the figure, $\angle ADE = \angle ACB$. Find x .



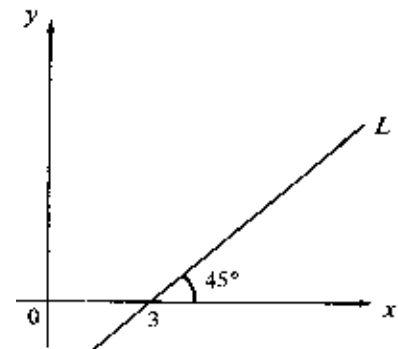
- A. 4
 B. 8
 C. 10
 D. 12
 E. 16

- 24 In the figure, $ABCD$ is a cyclic quadrilateral. If $\angle DAB = 110^\circ$ and $BC = BD$, find $\angle DAC$.

- A. 20°
 B. 35°
 C. 40°
 D. 55°
 E. 70°



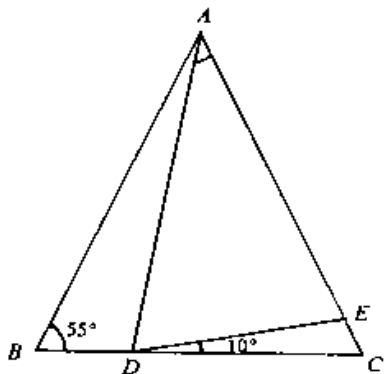
- 27 In the figure, the equation of the straight line L is



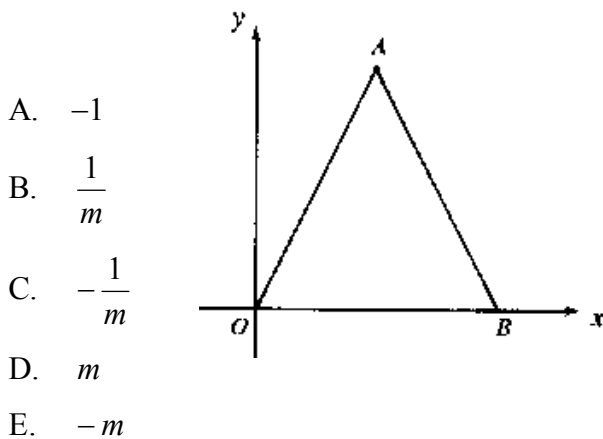
- A. $x - 3 = 0$
 B. $x - y - 3 = 0$
 C. $x - y + 3 = 0$
 D. $x + y - 3 = 0$
 E. $x + y + 3 = 0$

- 25 In the figure, $AB = AC$ and $AD = AE$. $\angle DAC =$

- A. 45°
 B. 50°
 C. 55°
 D. 60°
 E. 65°



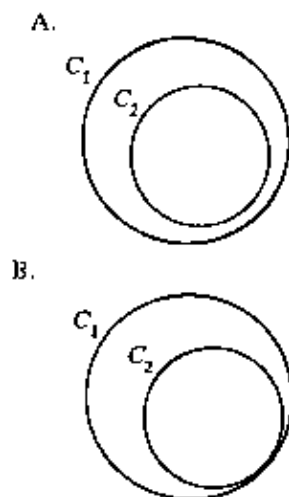
- 28 In the figure, $OA=AB$. If the slope of AB is m , find the slope of OA .



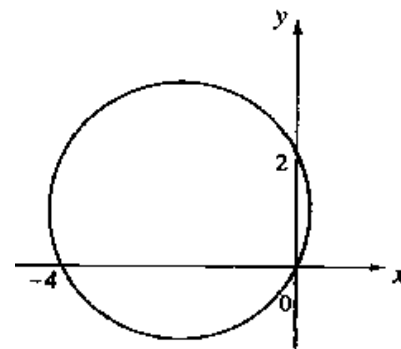
- 29 The table below shows the centers and radii of two circles C_1 and C_2 .

	Center	Radius
C_1	(2,2)	3
C_2	(5,-2)	2

Which of the following may represent the relative positions of C_1 and C_2 ?



- 30 In the figure, the equation of the circle is

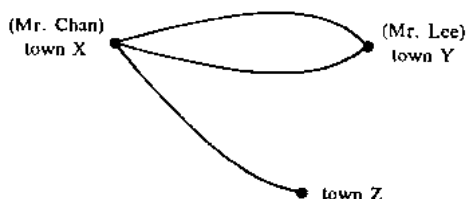


- A. $x^2 + y^2 - 5 = 0$
 B. $x^2 + y^2 - 2x + y = 0$
 C. $x^2 + y^2 + 2x - y = 0$
 D. $x^2 + y^2 - 4x + 2y = 0$
 E. $x^2 + y^2 + 4x - 2y = 0$
- 31 In a shooting game, the probability that A will hit

a target is $\frac{3}{5}$ and the probability that B will hit it is $\frac{2}{3}$. If each fires once, what is the probability that they will both miss the target?

- A. $\frac{1}{3}$
 B. $\frac{1}{4}$ D. $\frac{2}{15}$
 C. $\frac{2}{5}$ E. $\frac{11}{15}$

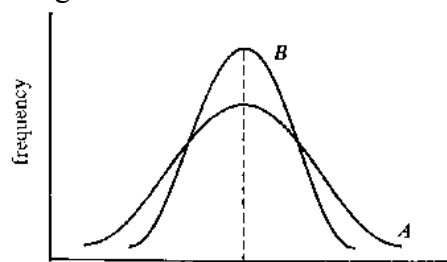
- 32 The figure shows that Mr. Chan has 3 ways to leave town X and Mr. Lee has 2 ways to leave town Y. Mr. Chan and Mr. Lee leave town X and town Y respectively at the same time. If they select their ways randomly, find the probability that they will meet on their way.



- A. $\frac{1}{2}$
 B. $\frac{1}{3}$ D. $\frac{1}{6}$
 C. $\frac{2}{3}$ E. $\frac{5}{6}$

- 33 The mean of a set of 9 numbers is 12. If the mean of the first 5 numbers is 8, the mean of the other four numbers is
- A. 4
 B. 10 D. 17
 C. 16 E. 25

- 34 The figure shows the frequency curves of two symmetric distributions A and B . Which of the following is/are true?



- I. The mean of A = the mean of B .
 II. The inter-quartile range of A > the inter-quartile range of B .
 III. The standard deviation of A > the standard deviation of B .

- A. I only.
 B. I and II only
 C. I and III only
 D. II and III only
 E. I, II and III

- 35 If $f(x) = \frac{x}{1-x}$, then $f\left(\frac{1}{x}\right)f(-x) =$

- A. $-\frac{1}{2}$
 B. -1 D. $\frac{x}{1-x^2}$
 C. $-\frac{1-x}{1+x}$ E. $\frac{x}{x^2-1}$

- 36 Factorize $2a^{n+1} - 7a^n - 30a^{n-1}$.

- A. $(a^n - 6)(2a + 5)$
 B. $a^n(a + 6)(2a - 5)$
 C. $a^n(a - 6)(2a + 5)$
 D. $a^{n-1}(a + 6)(2a - 5)$

E. $a^{n-1}(a-6)(2a+5)$

37

Simplify $\frac{\left(\frac{y}{x}-1\right)\left(1-\frac{x}{y}\right)}{\frac{x}{y}-\frac{y}{x}}$.

A. $\frac{x-y}{x+y}$

B. $-\frac{x-y}{x+y}$

C. $\frac{x+y}{x-y}$

D. $-\frac{x+y}{x-y}$

E. -1

38 If $5^a = 2^b = 10^c$ and a, b, c are non-zero,

then $\frac{c}{a} + \frac{c}{b} =$

A. $\frac{7}{10}$

B. 1

D. $\log 7$

C. 7

E. $\frac{1}{\log 2} + \frac{1}{\log 5}$

39 If α, β are the roots of the equation

$$x^2 - 4x - 3 = 0, \text{ then } \alpha^2 + \alpha\beta + \beta^2 =$$

A. -13

B. 5

D. 16

C. 13

E. 19

40 Find the range of values of k such that the

$$\text{equation } x^2 + (k-2)x + 1 = 0 \text{ has real roots.}$$

A. $k = 4$

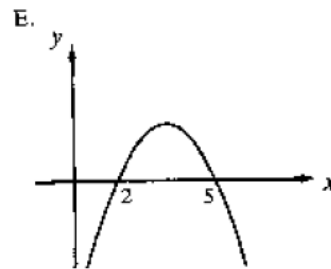
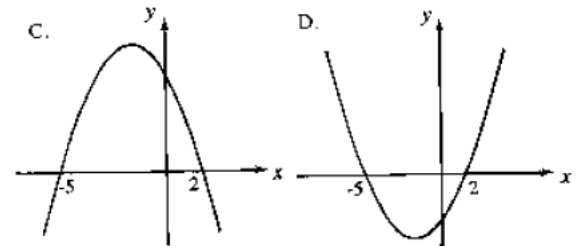
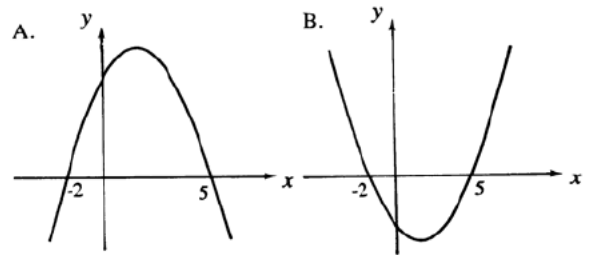
B. $0 < k < 4$

D. $k < 0 \text{ or } k > 4$

C. $0 \leq k \leq 4$

E. $k \leq 0 \text{ or } k \geq 4$

41 Which of the following may represent the graph of $y = -x^2 + 3x + 10$.



42 The first term of an arithmetic progression is 3 and the sum of the first 3 terms is 2. The common difference is

A. $-\frac{5}{3}$

B. -1

D. $\frac{5}{3}$

C. 1

E. $\frac{7}{3}$

43 If the geometric mean of two positive numbers a and b is 10, then $\log a + \log b =$

A. $\frac{1}{2}$

B. 1

D. 10

C. 2

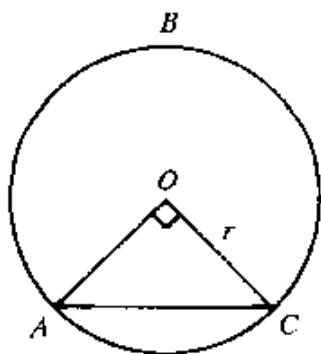
E. 100

44 The marked price of a toy is \$120 and the percentage profit is 60%. If the toy is sold at a discount of 20%, the profit is

- A. \$14.40
 B. \$21.00 D. \$33.60
 C. \$24.00 E. \$48.00

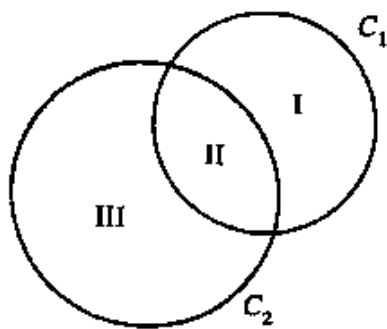
45 In the figure, O is the center of the circle. Find the area of the major segment ABC .

- A. $\frac{\pi}{4}r^2$
 B. $\frac{3\pi}{4}r^2$
 C. $\left(\frac{\pi}{4} - \frac{1}{2}\right)r^2$
 D. $\left(\frac{3\pi}{4} - \frac{1}{2}\right)r^2$
 E. $\left(\frac{3\pi}{4} + \frac{1}{2}\right)r^2$



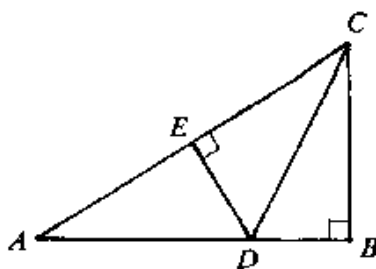
46 In the figure, C_1 and C_2 are two circles. If area of region I : area of region II : area of region III = 2 : 1 : 3, then radius of C_1 : radius C_2 =

- A. 9 : 16
 B. 2 : 3
 C. 3 : 4
 D. $\sqrt{2} : \sqrt{3}$
 E. $\sqrt{3} : 2$

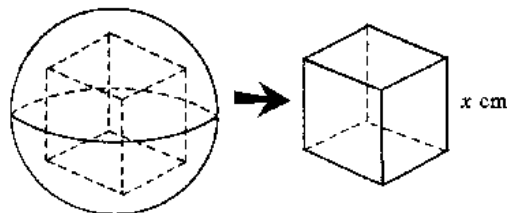


47 In the figure, $DE = DB$, $AC = 13$ and $BC = 5$. Area of $\triangle ADE$: Area of $\triangle ACB$ =

- A. 64 : 169
 B. 5 : 13
 C. 4 : 9
 D. 8 : 13



48 In the figure, a solid wooden sphere of radius 3cm is to be cut into a cube of side x cm. Find the largest possible value of x .



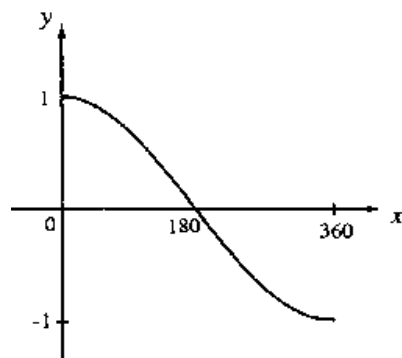
- A. $3\sqrt{2}$
 B. $2\sqrt{3}$ D. $\frac{3}{2}\sqrt{2}$
 C. 3 E. $\sqrt{3}$

49 If $0^\circ \leq x \leq 360^\circ$, the number of points of intersection of the graphs $y = \sin x$ and $y = \tan x$ is

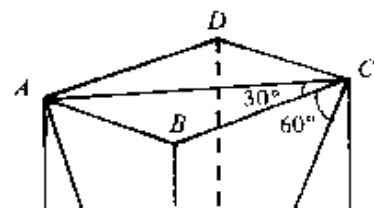
- A. 1
 B. 2 D. 4
 C. 3 E. 5

50 The figure shows the graph of the function

- A. $y = \cos \frac{x^\circ}{2}$
 B. $y = \frac{1}{2} \cos x^\circ$
 C. $y = \cos x^\circ$
 D. $y = 2 \cos x^\circ$
 E. $y = \cos 2x^\circ$



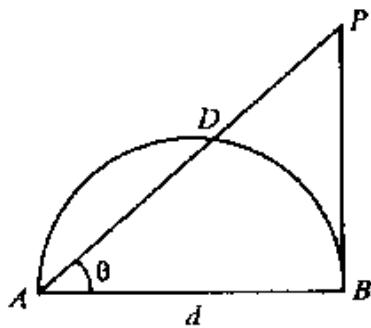
51 In the figure, $ABCDEFGH$ is a cuboid. $\tan \theta =$



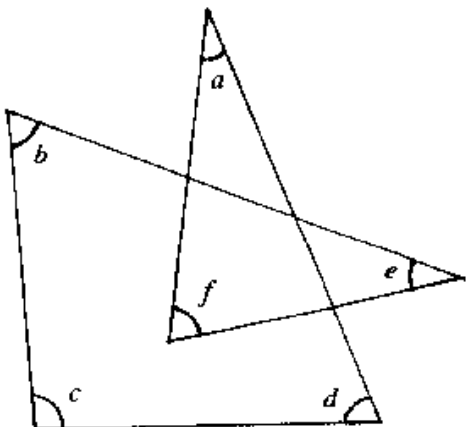
- A. $\frac{1}{3}$
 B. $\frac{1}{\sqrt{3}}$
 C. 1
 D. $\sqrt{3}$
 E. 3

- 52 In the figure, PB touches the semicircle ADB at B . $PD =$

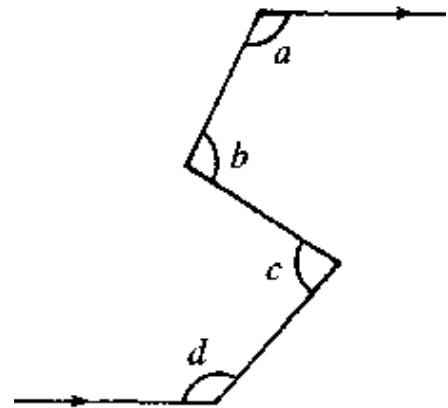
- A. $\frac{d}{2 \cos \theta}$
 B. $d \sin \theta \tan \theta$
 C. $\frac{d}{\sin \theta \tan \theta}$
 D. $\frac{d \cos \theta}{\tan \theta}$
 E. $\frac{d \tan \theta}{\cos \theta}$



- 53 In the figure, $a + b + c + d + e + f =$



- 54 According to the figure, which of the following must be true?



- A. $a + b = c + d$
 B. $a + d = b + c$
 C. $a + b + c + d = 360^\circ$
 D. $a + b + c + d = 540^\circ$
 E. $2a + 2b - c - d = 720^\circ$